**<u>Rethinking the Exit Velocity-Launch Angle Revolution in Baseball</u>**

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One of the most important questions for a Scout address in evaluating a College or High School Hitter is to predict whether they will be able to hit against higher velocity fastballs at the professional level. This assessment has become much more accurate over time as more College Parks across the country gain access to Batted Ball Data from technological advancements, such as TrackMan. However, there are still many environments where access to exit velocity and launch angle is not available. Thus, Amateur Scouts are forced to rely on pure observation without metrics to back up the "Eye Test". Without data to evaluate a hitter's performance, it can be riskier for Major League Organizations to draft a High School player. Oftentimes, teams will draft a College Player that is older and easier to predict future performance over a High School player with limited information but could potentially have more upside, if put into the right Player Development hands. However, there are certain observational attributes of a hitter's swing that scouts might be able to look for in a hitter that correlate with future performance versus 95+ mph Fastballs:

The main known indicators of success against Fastball Velocity are Bat Speed and Launch Angle. Pairing fast hands to catch up with higher velocities with a steep enough angle to get on plane with the ball has been the key methodology for many hitters nowadays. However, just like with pitching mechanics, no two hitter's swings are alike, and it is the job of the Hitter and Player Development Staff to find the ideal swing path for each individual player.

In this study, I collected data from *Baseball Savant* on various Swing Design, Bat Path, and Batted Ball metrics, modeling how they're associated with success against Fastballs:

| | Run Value | Expected On-Base | Hard Hit Rate | Whiff % |
|---|---|---|---|---|
| (Intercept) | 30.716 | 0.313 | -97.108*** | -68.212** |
| | (28.851) | (0.190) | (16.966) | (28.208) |
| Avg_Bat_Speed | -0.412 | 0.001 | 2.184*** | 1.269*** |
| | (0.429) | (0.003) | (0.235) | (0.420) |
| Hard_Swing_Percentage | 13.791** | 0.067* | | -0.377 |
| | (5.554) | (0.037) | | (5.430) |
| Swing_Length | -1.250 | -0.010 | -2.852* | -1.268 |
| | (1.323) | (0.009) | (1.716) | (1.293) |
| Swing_Tilt | 0.243** | 0.001 | -0.047 | 0.073 |
| | (0.118) | (0.001) | (0.151) | (0.115) |
| Attack_Angle | 0.262* | 0.003*** | 0.478*** | 0.282** |
| | (0.143) | (0.001) | (0.178) | (0.139) |
| Attack_Direction | -0.151 | 0.000 | 0.434*** | 0.449*** |
| | (0.126) | (0.001) | (0.164) | (0.123) |
| Contact_Point_vs_Batter | -0.183 | -0.001 | 0.158 | -0.031 |
| | (0.144) | (0.001) | (0.188) | (0.141) |
| Num.Obs. | 226 | 226 | 226 | 226 |
| R2 | 0.120 | 0.179 | 0.356 | 0.268 |
| R2 Adj. | 0.092 | 0.152 | 0.338 | 0.244 |
| AIC | 1455.6 | -815.5 | 1576.1 | 1445.4 |
| BIC | 1486.4 | -784.7 | 1603.5 | 1476.2 |
| Log.Lik. | -718.822 | 416.747 | -780.060 | -713.722 |
| F | 4.266 | 6.769 | 20.186 | 11.379 |
| RMSE | 5.82 | 0.04 | 7.63 | 5.69 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Key Definitions (according to *Baseball Savant* )

| | |
|---|---|
| **Attack Angle** | The vertical direction that the sweet spot of the bat is traveling at the moment it hits the baseball, as compared to the ground (Between 5-20 degrees is cosidered "ideal") |
| **Attack Direction** | The horizontal direction that the sweet spot of the bat is moving at the point of contact with the baseball (or the point the bat and ball cross paths on a swing-and-miss) |
| **Contact Point vs. Batter** | The "average" point at which their bat is nearest to the baseball on a swing (whether or not they make contact), measured relative to the batter's center of mass. This is also known as "Intercept Point". |
| **Swing Length** | The sum distance traveled by the head of the bat in XYZ space from the start of data until contact point (MLB League Average ~ 7.3 ft.) |
| **Swing Tilt** | A metric that measures the angular orientation of the "plane" of the swing and tells you the shape of a hitter's swing on the way toward contact. The tilt of the swing is defined as the vertical angle formed by the bat path compared to the ground. A higher angle is a "steeper" swing (closer to 90 degrees) and a lower angle is a "flatter" swing (closer to 0 degrees). |

Each Regression model is based on a key metric versus Fastballs and Sinkers: Run Value (total runs created or diminished versus a specific pitch); Average Expected On-Base (or "wOBA") calculated based on a hitter's exit velocity and launch angle; Hard Hit Percentage on all Batted Ball Events; and Whiff Percentage (the total number of swing-and-misses divided by the total amount of a hitter's swings).

None of the model's combinations of these variables were highly correlated with fastball success, but this could be a by-product of the variability of swing design and a hitter's plate discipline and approach. The highest R-Squared being 0.356 (i.e. only ~ 30% of the model actually explains how hard contact is produced against fastballs) is a testament to that statement. However, that being said there were some takeaways:

What has been said remains to be true: Bat Speed and Launch Angle are still the key performance indicators versus velocity. However, one variable that was found to be statistically significant stood out: Swing Length. **According to this model, hitters that have a swing that is a foot longer, on average compared to the league, make approximately 3% less hard contact against fastballs than hitters with shorter or average swing length.** This may seem counterintuitive to some in the industry, as studies have shown Bat Speed and Swing Length to be correlated, since bat speed generally increases throughout the length of the swing. Many hitting coaches try to teach players to keep the barrel of the bat through the zone for as long as possible, but the reality is that swing design is a complicated series of levers. What works for one player will not always work for another. For about a while now, hitters have been told to not only launch the ball, but to pull the ball in the air (henceforth, getting the bat head further out in front) and swing harder to produce extra-base hits at the expense of higher strikeout rates. So how does *swing length* factor into this?

| 2025 Hard Hit % vs. FB (League Avg. = 47.4) | | Swing Length (ft.) | | | |
|---|---|---|---|---|---|
| | | Short | Average | Long | AVG |
| Attack Angle | Flat | 33.1 | 47.1 | N/A | 42 |
| | Average | 45.8 | 47.7 | 50.3 | 47.7 |
| | Steep | N/A | 50.7 | 46.1 | 49.6 |
| | AVG | 41.9 | 48.1 | 48.7 | 47.4 |

In 2026 thus far, swing lengths have varied from as long as almost 9 feet in length (Junior Caminero- 8.7 feet) and as short as almost half of that (Luis Arraez- 5.9 feet). When comparing hitters' attack angle versus swing length, the data shows that, generally, that the trend of hitters being able to create backspin on the ball through creating launch in their swing and, thus, getting on plane with the ball tend to create harder contact holds true. This statement remains the same versus fastball velocity and is why the industry continues to pursue hitters that pair launch angle with bat speed. Assuming hitters with league average attack angle, **longer swings, on average, generate nearly 3% more hard contact than average length swings (50.3% versus 47.7%)**.

However, there is an interesting anecdote to this: while longer swings tend to have more success versus fastballs, there can be instances where it hits its breaking point. **Hitters that have steeper attack angles generate, on average, 4% more hard contact <u>when they have an average swing length</u> versus a longer swing length (50.7% versus 46.1%)**. Thinking about this logically, it makes sense that hitters would have a tough time versus 95+ mph when the barrel of the bat takes a longer path towards its point of contact, especially in a day in age where pitchers attack the top of the zone with velocity. Therefore, there can be cases where the pairing of steeper launch angle with a longer swing length hits its ceiling. While many organizations have accepted the fact that hitters with heavy launch are going to strike out at a higher rate, it becomes a different story when it also results in diminished hard contact.

Take **Nolan Gorman** for example, who has a higher-than-average attack angle of 17 degrees (league average is 10 degrees) and a longer swing averaging ~ 7.6 feet. Gorman, a former first round pick that was recently demoted back down to AAA after struggles this season, has **some of the worst chase and strikeout rates in the league** (1st and 4th percentile rankings respectively); however, **the hard contact is sparse**, only ranking in the 65th percentile in Hard Contact Rate versus all pitches (43.8 %). Gorman definitely leans closer to the extreme ends of these spectrums and is potentially a candidate who could afford to change up his mechanics. Whether that means being shorter to the ball by getting the barrel more out in front and generating a flatter swing or letting the ball travel deeper in the zone, Gorman has plenty of raw power that might be worth sacrificing a bit for better contact rates.

**Junior Caminero** poses as an example of an outlier: the hitter with the longest average swing length in the Majors (8.7 ft.) and, arguably, a top candidate for AL MVP. Caminero absolutely demolishes fastballs with a combined Run Value of 18 versus Fastballs and Sinkers, in large part due to his elite bat speed levels, averaging just under 80 mph (79.8 mph). However, unlike a lot of power hitters in the game who get under the ball, **his attack angle is on the flatter side averaging around 8 degrees** but is just enough in the ideal zone to allow him to get on plane with fastballs, especially those thrown in the top of the zone. Surely a good deal of his ability to generate electric bat speed and stay through the zone as long as possible has to do with his larger-than-life 220 lb. frame and long limbs. He might not always get on plane with the ball and get the barrel on the sweet spot, but he makes up for it with physicality and bat speed while staying through the zone for as long as possible that causes him to absolutely punish fastballs.

Rather than being glued to the end result exit velocity off the bat, it might be helpful for scouts to think about bat speed in a different way: How long or short is the batter's swing? Does his swing stay through the zone or his approach short to the ball and get the barrel out in front? What swing design would be optimal based on the type of hitter he is?

Recent studies have been conducted from the relatively new contact point data provided from Baseball Savant and how far out in front of a hitter's center of mass should they be to produce hard contact. However, perhaps a study that would be worth looking into would not only show how far out in front the batter tends to make contact but also *when* they're starting their swing. Furthermore, combining this with a hitter's point of contact, attack angle, and length of swing, how do these variables interact with one another that affect the outcome of a batted ball?

There are many variables when considering a pitcher's pitch design and how to best optimize their profile based on their size, mechanics, and arsenal. As hitting technology continues to advance, we may be approaching a day where we start looking at hitters through the same lens.